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Development of a Machine Learning Driven Solution to Manage Sulphate Scaling in a Giant Field, Offshore Abu Dhabi

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Abstract

Mineral scale formation, particularly strontium sulphate (Celestite), poses a challenge in ADNOC offshore oilfield operations. Seawater injection, while effective for reservoir pressure maintenance, introduces scaling risks due to the mixing of incompatible waters—formation water rich in divalent cations and seawater containing sulphate ions—leading to precipitation of sparingly soluble salts. The framework integrates machine learning models trained on sparse historical and synthetic datasets to estimate seawater breakthrough, reconstruct ion concentrations, and predict Celestite Saturation Index (SI) without full thermodynamic simulations. Artificial neural networks were employed, achieving seawater percentage prediction accuracies of 95.8% and 96.3% for two reservoirs, outperforming traditional operator formulas. The Gaussian process regression for ion concentration reconstruction using specific gravity as input yielded high accuracies (above 95%) for most ions, with acceptable performance for strontium (88.21%) and sulphate (84.53%). The scaling index (SI) prediction using classic machine learning model algorithms achieved predictions within $\pm 0.05 \log_{10}$ units in the critical "watch" zone ($SI > 0$). The system is deployed via a large language model interface, enabling natural language queries to retrieve time series data, documents, and trigger model predictions. This workflow streamlines scale monitoring and decision-making offering a scalable, traceable, and responsive alternative to traditional workflows in offshore scale management.

Introduction

The efficient management of oilfield operations hinges on proactive strategies that mitigate potential disruptions such as mineral scale formation [1]. Scaling, the precipitation and deposition of inorganic salts poses a significant challenge to production, flow assurance, and the integrity of subsurface and surface infrastructure. These mineral deposits can restrict fluid flow, damage equipment, and necessitate costly interventions, ultimately impacting the economic viability of oil and gas extraction [1]. Seawater injection is a widely employed technique aimed at maintaining reservoir pressure and sweeping oil towards producing wells. In offshore fields, seawater is a common choice for injection due to its availability and relatively low cost. While effective in bolstering oil production, seawater injection introduces a critical challenge:

the potential for inorganic scaling [2]. This phenomenon arises from the commingling of incompatible waters within the reservoir and production system [3]. Formation water, naturally present in the reservoir, typically contains high concentrations of divalent cations such as barium (Ba^{2+}), strontium (Sr^{2+}), and calcium (Ca^{2+}) ions. Injected seawater is instead rich in carbonate and sulphate ions (SO_4^{2-}): when these two chemically distinct water sources mix, the ionic compositions can lead to supersaturation and the subsequent precipitation of sparingly soluble salts such as sulphate (BaSO_4), strontium sulphate (SrSO_4), and calcium sulphate (CaSO_4) [2] [4] [5]. These precipitates, particularly in the form of crystalline structures like barite (BaSO_4) and celestite (SrSO_4), can severely restrict flow pathways in porous media near water injection wells and oil producer wells, impair well injectivity and productivity, and necessitate costly remediation interventions. Strontium sulphate is often thermodynamically more stable than barium sulphate under certain conditions and tends to precipitate as celestite within well tubing and production facilities [6].

The formation of sulphate scale presents significant operational and economic burdens, leading to reduced wellbore diameter and permeability impairment, equipment damage, production deferment, and increased operational expenditure due to the high cost of chemical treatments, mechanical interventions, and lost production [1] [7]. The removal of sulphate scale from offshore wells is particularly difficult and costly. Both chemical and mechanical techniques, such as milling, high-pressure water jetting, sandblasting, and chemical chelates/solvents, can be used to remove the scales. Chemical methods, often used prior to mechanical treatment, are generally cheaper and more easily applicable to scales that are not easily accessible (such as SrSO_4) or formed under high-temperature and high-pressure conditions. The current scale risk evaluation routines heavily depend on water analysis from samples collected offshore and later examined in specialized laboratories where the ratio between seawater and formation water is subsequently inferred via empirical models and physical simulators. These methods are prone to significant delays and possess limited accuracy. Also, evaluations based on chemistry alone often fail to account for crucial real-time operational parameters such as well intervention history, flow conditions, and dynamic changes in the subsurface environment. For instance, thermodynamic modeling, while indicating whether minerals are undersaturated or supersaturated, often falls short in predicting actual scale deposition because it does not fully account for kinetic factors such as metastability and the influence of fluid flow dynamics [8]. Furthermore, the impact of hydrodynamics, including turbulence and wall shear stress, significantly affects scale formation and deposition location, often promoting scaling in areas that would otherwise remain metastable under laminar conditions [8].

This work introduces an AI-agentic framework designed to aid strontium sulphate (Celestite) scale risk management. Our solution overcomes the limitations of conventional approaches by leveraging incomplete chemical analysis or even cases with no water analysis, along with real-time sensor data and engineering heuristics to dynamically assess scaling risk. The core of our approach lies in a machine learning framework integrated with a domain-specific data model allowing the contextualization of production data from sensors, with well integrity philosophy documents and laboratory analysis. This modular, data-centric architecture enables traceable, scalable, and real-time insights through agentic-AI interfaces and makes the solution adaptable to other oil and gas fields facing similar scale risk challenges under variable waterflooding conditions. The framework consists of machine learning models that are trained on sparse historical water chemistry data addressing three problems. The first aim is to predict the seawater percentage at the wellbore from produced water specific gravity data for a given reservoir and formation, known to be a readily available proxy for seawater breakthrough [9]. Similarly, the second model uses the produced water specific gravity to estimate the concentration of selected chemical species ($[\text{Sr}^{2+}]$, $[\text{SO}_4^{2-}]$, $[\text{Cl}^-]$, etc., required for the computation of the scaling index with a simulation software. Subsequently, the whole simulation workflow was surrogated by a machine learning model, bypassing the need for full thermodynamic simulations for SI estimation.

Generative AI, LLM-based agents were then deployed as a user interface for simplifying scale management operations, serving production data, results from ML models, and operational guidance from

documents through a single interface. This AI-agentic system has the potential to reduce the reliance on detailed laboratory analyses and complex simulations, facilitating timely and more confident decision-making regarding scale management.

Finally, flow dynamics simulations were integrated with field data in the attempt to enhance the SI prediction by including features going beyond mineral deposition thermodynamics. Such features include both static (well trajectory, completion, etc.) and fluid dynamics (Reynolds number, liquid slugging, kinematic viscosity, etc.) features, in an attempt to improve the accuracy of the SI prediction. We found this approach to be promising, although realistic SI estimations are seldom obtained given the lack of data on the progressive deposition and growth of scale through time.

Architecture and Data Modelling

Agentic artificial intelligence (AI) systems represent a transformative evolution from reactive, narrowly focused models to autonomous agents capable of multi-step planning, dynamic reasoning, and adaptive interactions with their environment. The core premise enabling these advanced functionalities lies in the quality of the data used to train, update, and guide these systems.

Agentic AI requires data to be modeled appropriately because the quality, structure, and contextual integrity of the data fundamentally determine the system's ability to perceive its environment, reason effectively, plan autonomously, and adapt to unforeseen circumstances. Inadequate data modeling introduces risks that range from algorithmic bias and emergent unsafe behaviors to a breakdown in multi-agent coordination and systemic vulnerabilities. By contrast, appropriately modeled data is foundational to ensuring that agentic AI behaves reliably, safely, and robustly in complex and unpredictable operational contexts. Inaccuracies, biases, or inadequacies in data modeling can compromise system performance by leading to misinterpretation of inputs, erroneous internal representations, and ultimately unsafe or suboptimal decision-making [10].

Data from production facilities consisting of well rates, pressure and temperature measurements, and water chemistry were integrated with well schematics, stimulations jobs records and scale squeeze treatments in a domain specific knowledge graph. Time series data were transformed to a set structure (format, unit of measurement, etc.), while other "static" data were extracted from documents and completion by parsing PDF files. A hierarchical structure was then built to represent the relationships between each type of data and its provenance, as shown in [Figure 1](#).

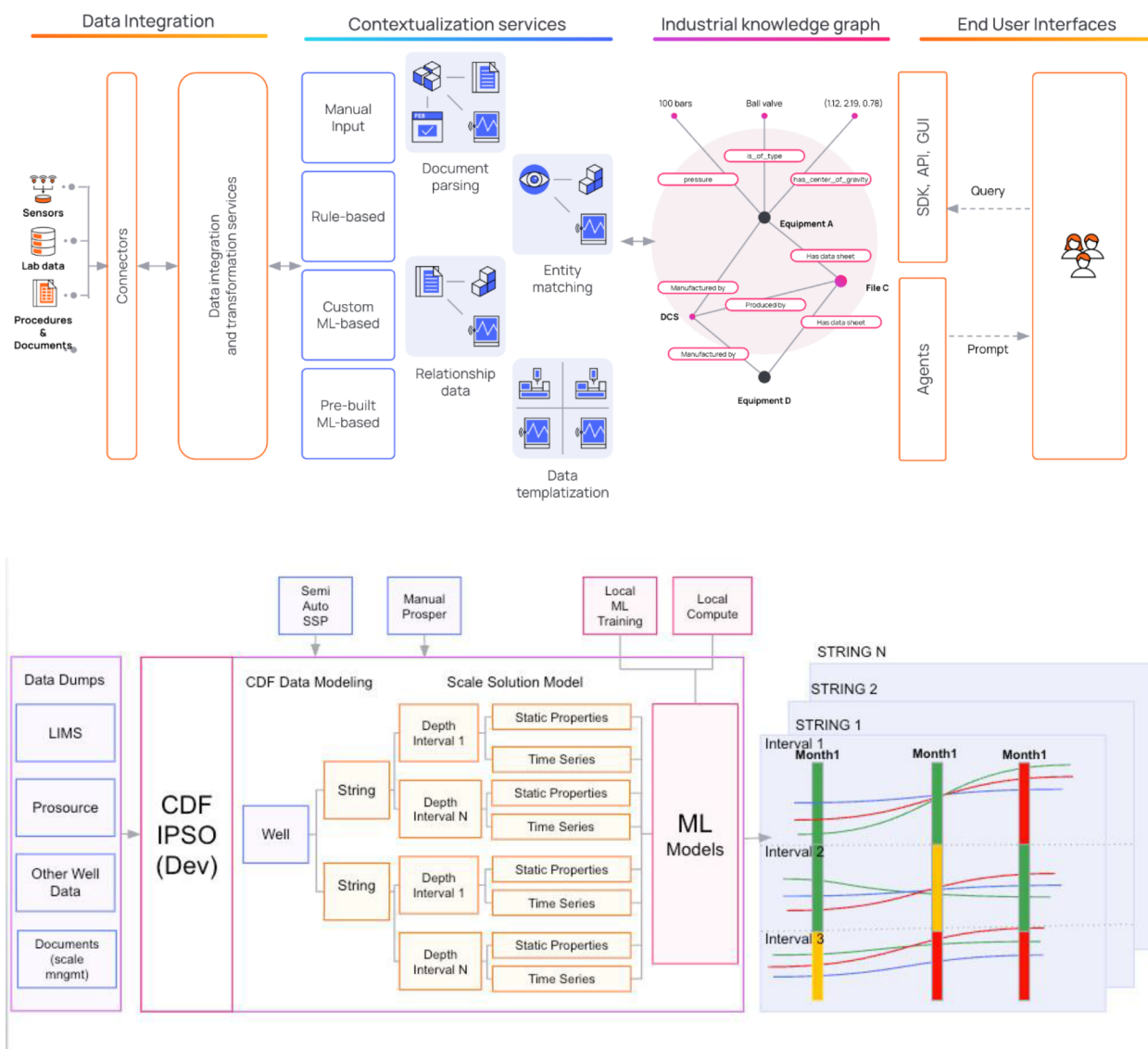


Figure 1—Overview of the data flow from integration and contextualization to an industrial knowledge graph, enabling end-user interactions through intelligent agents.

The architecture of the proposed AI-driven solution is composed of several layers including a cloud historical layer, a data processing and feature engineering layer as a data platform. The data platform is then leveraged for the development of data science machine learning algorithms for inferring produced water chemistry properties. Also, to operate generative AI agents which can access the data science models, operational data and document metadata, enabling a comprehensive understanding of the context. AI agents use language models (from OpenAI, Anthropic, Mistral and Google) to interpret prompts with instructions and goals, build industry-relevant tools, and access to industrial data stored in the knowledge graph. Strontium sulfate (Celestite) scaling risk is commonly assessed by calculating the Scaling Tendency (ST) or the Scaling Index (SI) defined as follows for Celestite:

$$\begin{aligned}
 \text{Sr}^{2+} + \text{SO}_4^{2-} &\rightleftharpoons \text{SrSO}_4 \downarrow \\
 K_{sp}(\text{SrSO}_4) &= [\text{Sr}^{2+}] \cdot [\text{SO}_4^{2-}] \\
 ST(\text{SrSO}_4) &= \frac{a_{\text{Sr}^{2+}} \cdot a_{\text{SO}_4^{2-}}}{K_{sp}(\text{SrSO}_4)} \quad SI(\text{SrSO}_4) = \log[ST(\text{SrSO}_4)]
 \end{aligned}$$

Equation 1—Celestite (SrSO_4) equilibrium reaction and definitions of the celestite scaling tendency (ST) and scaling index (SI). The two rely on the chemical activity of the aqueous species and their solubility product (K_{sp}).

Where square brackets represent ionic concentrations at equilibrium, K_{sp} the solubility product and a the chemical activity coefficient for a given species. Note that (from the thermodynamic point of view) in this notation, $SI < 0$ indicates sub-saturation, and the solid is not expected to form, at $SI = 0$ the solid is in equilibrium with water, while at $SI > 0$ the solution is in supersaturation, and solids are expected to form.

Calculations are typically done by running thermodynamic simulators, which relies on electrolyte theory to calculate activity coefficients and equation-of-state models for oil and gas properties to compute activity coefficients for mineral species at given temperature and pressure conditions. These simulators require comprehensive water chemistry data, including specific gravity (SG) and major ion concentrations such as strontium (Sr^{2+}), sulfate (SO_4^{2-}), and chloride (Cl^-). However, two key challenges arise. First, complete water chemistry data are rarely available. For the offshore field studied, between 2017 and 2024, only 1–2 full water chemistry analyses per well were typically performed annually. Second, water analysis is time-consuming, particularly in large offshore operations with thousands of wells. Laboratory facilities are often distant from the field and constrained by sample throughput. Consequently, scaling risk assessment and related operational decisions are significantly delayed. In the absence of full water chemistry data, operators typically estimate scaling index by calculating the seawater fraction in the produced fluid and referencing pre-calculated scaling envelopes (Figure 2). These envelopes classify different seawater percentage (SW%) ranges into zones representing increasing scaling risk: equilibrium, scale precipitation, and scale deposition. Each scaling envelope is generated for specific temperature and operating pressure conditions by running scale prediction simulations, varying only the seawater fraction while assuming the formation water composition remains constant over time. The seawater percentage (SW%) is estimated using an empirical correlation, typically based on available measurements. In most cases, only specific gravity (SG) data are available

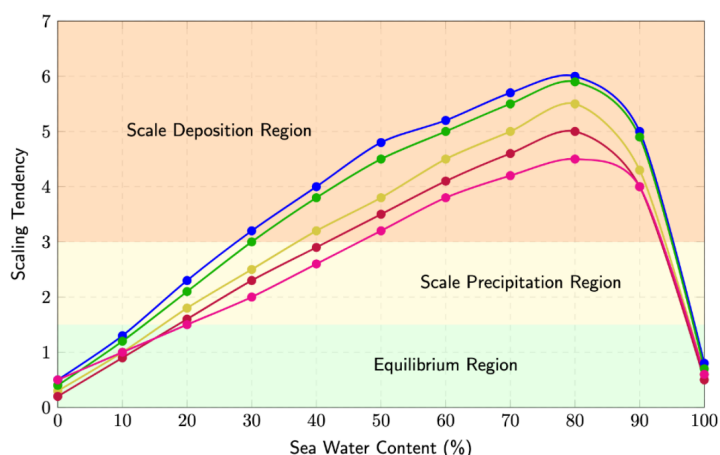


Figure 2—Scaling index envelopes as a function of seawater percentage (SW%) for several reservoirs (different color lines for different formation water chemical compositions) computed at 98°C and 200 Bar. Three distinct risk zones are identified as "Equilibrium Region" ($ST < 1.5$), "Scale Precipitation Region" ($1.5 < ST < 3$) and "Scale Deposition Region" ($ST > 3$).

$$SW\% = \frac{100}{4} \left(\frac{SG_{fw} - SG_{20^\circ C}}{SG_{fw} - SG_{sw}} + \frac{Sr_{fw} - Sr_{20^\circ C}}{Sr_{fw} - Sr_{sw}} + \frac{SO_{4fw} - SO_{420^\circ C}}{SO_{4fw} - SO_{4sw}} + \frac{Cl_{fw} - Cl_{20^\circ C}}{Cl_{fw} - Cl_{sw}} \right)$$

Equation 2—Operator's formula for estimating seawater percentage prediction. Where, SG refers to specific gravity, Sr denotes strontium concentration, SO_4 represents sulfate concentration, and Cl indicates chloride concentration. Subscripts are used to identify the water source, where fw stands for formation water and sw refers to sea water. The notation $20^\circ C$ specifies that the sample is measured at a temperature of $20^\circ C$.

In order to establish a fast and reliable method to infer SW% from readily available field data, a machine learning model was developed and trained on historical chemical data with the goal to improve the estimation accuracy over the analytical formula. Our approach, SG is the sole input parameter used during model training of an artificial neural network. Our model receives a single input (SG), followed by three successive pairs of dense and batch normalization layers. By integrating normalization directly into the model architecture, we mitigate internal covariate shift, leading to more stable and predictable gradient behavior [14]. Two separate models were trained, one for reservoir A another for reservoir B, and where the equation 1 was used as a ground truth for SW%, serving as the target for all the ML model training. The full four input parameters: specific gravity (SG), strontium (Sr), sulfate (SO_4^{2-}), and chloride (Cl) were obtained from chemical analysis spanning from the first recorded well data. The dataset was then split into 80% for model training and 20% testing. The training portion was further divided into 85% for model training and 15% for validation. All input features were normalized using a Min-Max scaler to prevent exploding gradients and to ensure that each parameter contributed proportionally to the model. Since the model has only one input parameter—specific gravity (SG)—and SG exhibits an approximately linear relationship with the seawater percentage, we use a linear unit [15] as the activation function. The output layer consists of a single-node dense layer, and the model is trained using the mean squared error (MSE) loss function. A complete overview of the model architecture is provided in figure 3 with a total of 2953 parameters.

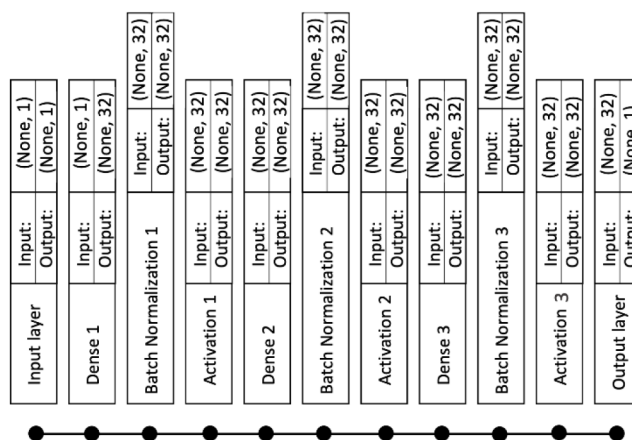


Figure 3—Model architecture for seawater percentage ML prediction.

Figure 4 presents a two-way comparison of seawater percentage (SW%) estimation. The blue dots represent the true SW% values plotted against specific gravity. Note that multiple SW% outcomes are possible for a given SG: this variability arises as different combinations of ion concentrations can produce the same SG. The green line shows the SW% estimated using the analytical formula [ref] with SG as the only input, while the red line represents the machine learning (ML) model's prediction, also using SG as the sole input.

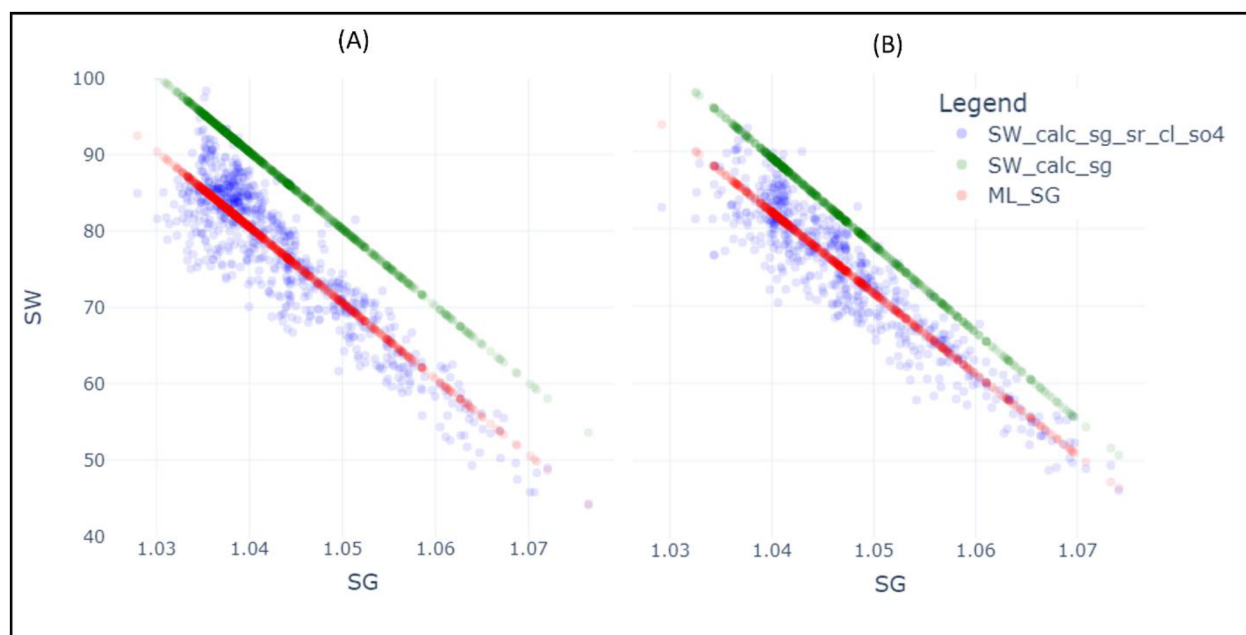


Figure 4—Comparison of seawater percentage (SW%) versus specific gravity (SG) for two reservoirs: (A) Reservoir A and (B) Reservoir B. The blue points represent the true SW% values. The green line corresponds to the analytical formula prediction, while the red line represents the machine learning (ML) model prediction using SG as the only input feature.

Table 1—Comparison of analytical and machine learning model accuracies across two reservoirs with corresponding test sample counts.

Reservoir	Analytical Model (Accuracy %)	Machine Learning Model (Accuracy %)	Test Sample Count
Reservoir A	86.80%	95.83%	754
Reservoir B	90.84%	96.30%	586

Both estimation methods follow the expected inverse trend between SG and SW%, indicating that the overall relationship is captured. However, the analytical formula, when used with SG alone, achieved an accuracy of only 86.8%, compared to 95.83% for the ML model. Notably, the analytical method tends to consistently overestimate seawater percentage. While the SW% provides a general indication of the extent of seawater mixing with formation water, the scale index is more directly informative in determining whether a well is experiencing scaling. In Figure 2, using a predefined scale envelope at 98 °C and 200 bar, an SG value of 1.07 yields SW% estimates of 60% from the ML model and 50% from the analytical formula. When these values are plotted on the scale envelope for the reservoir A, the corresponding scaling indices are 3.7 and 3.0, respectively. The analytical estimate places the point within the deposition region, while the ML estimate lies near the boundary with the precipitation zone. This example highlights that even small differences in SW% estimates can lead to different scaling assessments under current boundary conditions. Therefore, having an accurate SW% estimator is crucial for effectively tracking and managing well conditions.

Water Chemistry Reconstruction

The calculation of the scaling index is typically performed by a chemical simulator. This requires the knowledge of several physical water parameters such as water electrical resistivity, pH, temperature, pressure, and the individual concentrations of the major cations (Na⁺, K⁺, Ca²⁺, Mg²⁺, Sr²⁺, and Ba²⁺) and anions (Cl⁻, SO₄²⁻, HCO₃⁻, and CO₃²⁻), which can be obtained from a full chemical analysis of

produced water [12]. The scaling index quantifies the thermodynamic driving force for scale formation under specific conditions [13]. As the full water chemical analysis may not be available, a machine learning model was trained to predict physical and chemical water parameters from incomplete data and estimate the uncertainty associated with the prediction. A Gaussian process regression (GPR) algorithm was implemented to impute chemical data from incomplete chemical data. As a non-parametric Bayesian approach, GPR allows to model the absolute value of each quantity providing a measure of the inference's uncertainty and allowing an informed decision making.

A GPR model was trained using data from reservoir A, employing a composite kernel consisting of a Radial Basis Function (RBF) and a White Kernel [15] to capture smooth variations and observation noise. Only samples with complete chemical analysis were considered. Out of a total of 3,773 samples, 3,207 were used for training and 566 (85:15 split) for testing the GPR model. Table 2 presents the mean absolute percentage error (MAPE) of the GPR predictions for each ion. Lower MAPE values indicate higher prediction accuracy, with Cl and Salinity achieving the best results, while Mg and SO₄ show higher errors.

Table 2—Mean absolute percentage error (MAPE) and percentage accuracy for key input water chemistry parameters.

Input param	MAPE %	Accuracy (100% - MAPE)
Sr	11.79	88.21
SO4	15.47	84.53
Cl	2.04	97.96
Na	3.07	96.93
Mg	17.65	82.35
Salinity	2.74	97.26

The GPR results for most ions have an accuracy of 95% or above when inferring ion concentrations from SG only as input. The case of salinity (Figure 5.A), and similarly for other species such as chlorides, sodium, etc., shows a consistent accurate estimation across a large interval of SG as the relationship between the two is practically linear, making it easy to learn for the algorithm. In the case of Strontium (Figure 5.B) instead, the overall accuracy drops to 88% but it is apparent how this is not uniform across all the SG intervals. Indeed, the range between SG=1.035 and 1.050, corresponding to [Sr] of 50 mg/L and 110 mg/L respectively, is where most historical data cluster and here the model achieves an accuracy above 90%. This then drops to higher SG (lower SW%) where less training data is available. In contrast, for sulfate (Figure 5.C), the GPR prediction is less accurate across all SG values, reflecting a less linear relationship between these features.

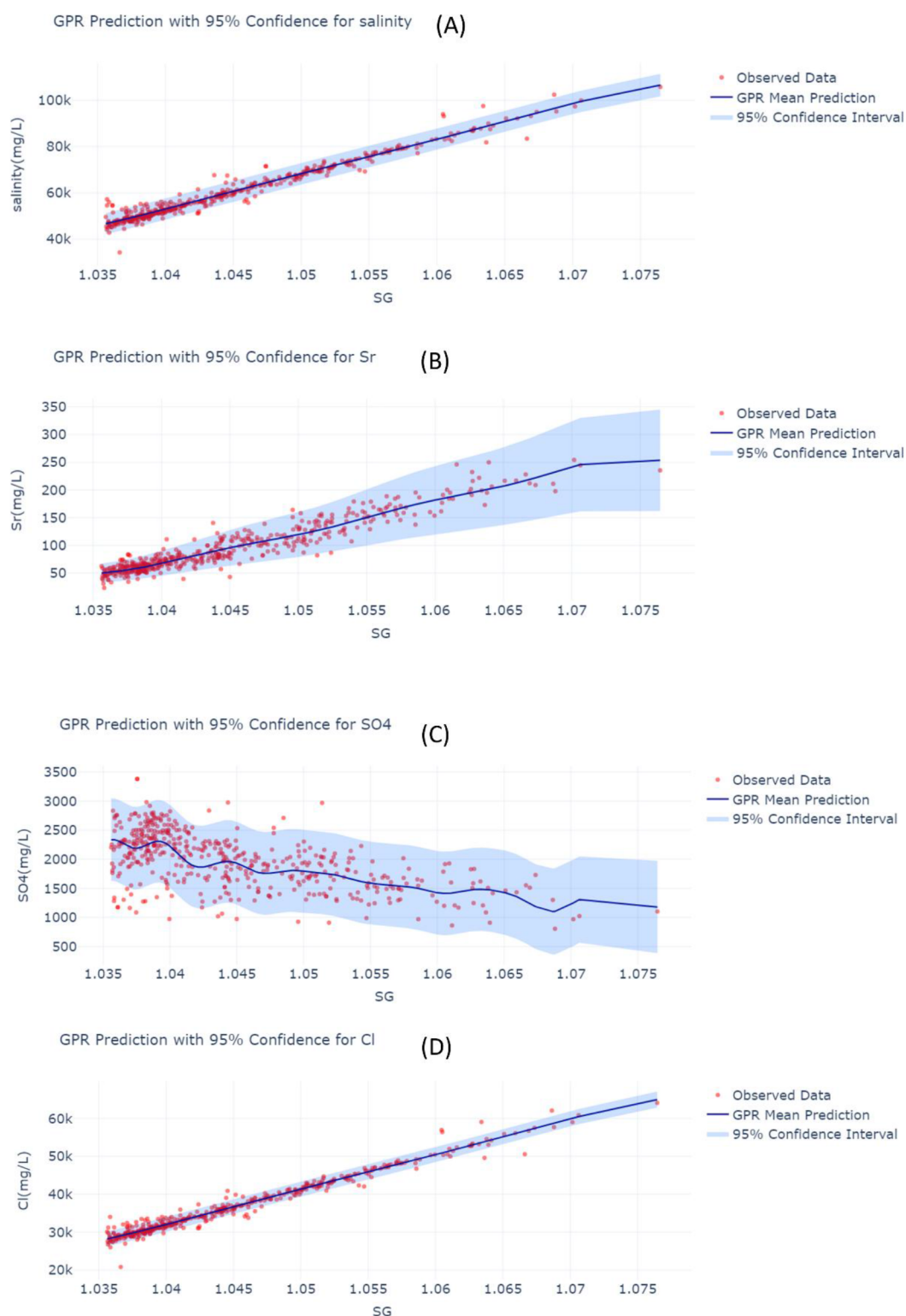


Figure 5—A–D – Gaussian Process Regression (GPR) predictions with 95% confidence intervals for reconstructed (A) salinity, (B) Sr, (C) SO₄, and (D) Cl as functions of specific gravity (SG). Red dots represent observed values, the solid blue line indicates the GPR mean prediction, and the shaded blue area denotes the 95% confidence interval.

To further improve the model accuracy, proximity-wise model training was also tested. For each well, a predetermined number of surrounding wells based on distance are used for training and evaluating it for that one well. The rationale behind it is that closer wells should have more identical properties, thus improving the model. Although the number of surrounding wells is fixed during training, there are few heterogeneities:

one is the number of training and evaluation samples. Throughout the lifecycle of each well, there is no fixed number of water chemistry analysis, thus some wells may have only one data point at the least or could have more than 20.

For this experiment, using 30 surrounding wells with the same only SG as input, same kernel and training configurations from the reservoir model. Figure 6 shows the results of the MAE for each ion. Compared with the reservoir-wise, lower MAPE can be achieved, however, this depends on the well. This variability introduces additional complexity when comparing wells, as some models benefit from richer training data while others suffer from sparse observations. Consequently, model performance is influenced not only by proximity but also by the distribution and quality of available samples.

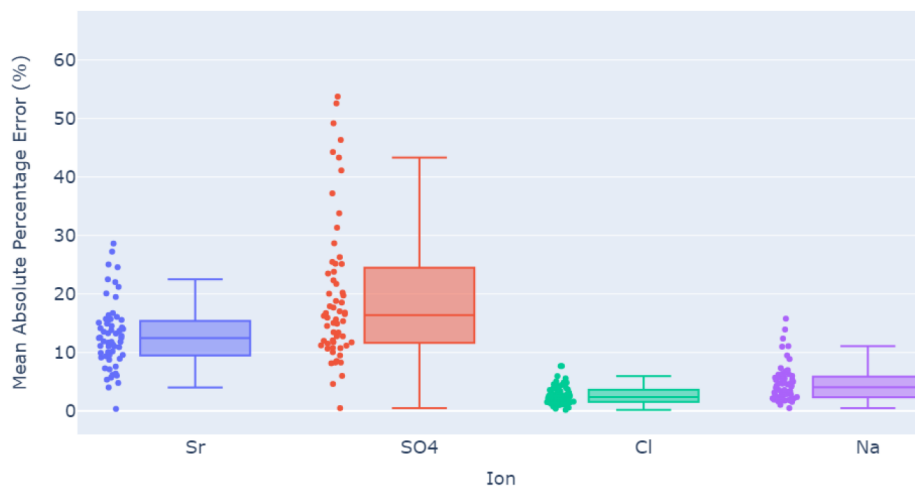


Figure 6—Mean Absolute Percentage Error (MAPE) of proximity-wise modeling for predicting ion concentrations across wells. Each dot represents one well.

Celestite SI inference from chemical data

In a similar fashion, to enable constant and topical celestite scale surveillance at scale, it should be possible to infer directly the Saturation Index / scaling index (SI) from sparse chemical data. Traditional SI calculation methods scale poorly and only provide point-in-time snapshots of chemical sulfate scaling risk, leaving gaps in overall visibility. A quick and easy way to compute SI from e.g. SG would be a tool to get an estimate of scaling index on demand. Also, scaling simulations require manual input of production and chemistry data, making the workflow hard to scale for a giant oilfield at the scale of thousands of wells. Machine learning allows for instant SI estimations from a limited amount of information – using water chemistry estimations of SO_4^{2-} and Sr^{2+} from SG – to provide users with an estimated Celestite scale risk envelope at a specific well condition.

The scale simulator workflow was first automatized in both its calibration and simulation mode to allow the exploration of parameters such as temperature, pressures, full chemical analysis, and production values. Then, the simulator was surrogated by running 35,000 total simulations, spanning 3500 laboratory analysis samples. This provides the machine learning model with the full history of produced water analysis for a reservoir – and exposes the model to all possible temperature and pressure combinations encountered along the well string and their impact on the SI. This use case closes the loop on estimating the full chemical risk for Celestite – translating estimated or definitive chemical analysis into a specific chemical risk at a position along a given well string (at a given temperature and pressure). A simple correlation analysis of and $[\text{SO}_4]$ concentration effects over SI, shows an almost linear correlation over a large span of concentrations, with exception of very low values, below 50 ppm (Figure 7). This was also observed for SG and temperature, providing confidence over the ability of machine learning techniques to learn this correlation and predict accurately SI from partial produced water chemical data. The strongest correlation

observed is when considering the logarithm of the molalities of [Sr] and [SO₄], where strong linearity is observed. Note that – while observing a favorable signal for SI prediction – it is assumed that all of the fluids deviation from pure water is attributable to [Sr] and [SO₄] mass fractions. We expect this to Bias the SI predictions upwards.

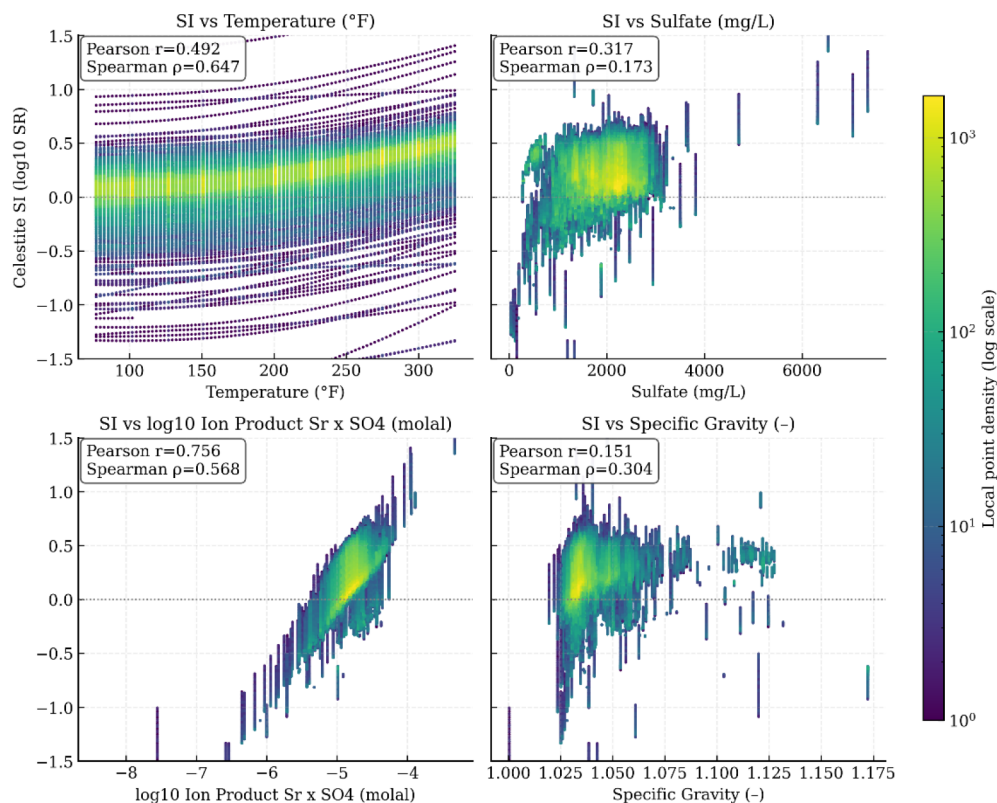


Figure 7—Feature correlation plots results for Celestite SI. Temperature (top-left), SO₄ concentration mg/L (top-right), log(Sr * SO₄) Molal (bottom left) and specific gravity (bottom-right)

The model selected to estimate Celestite SI from partial water chemistry data is a two-stage classification and regression pipeline. First, considering the practical realities of the scaling problem: we don't care nearly as much about the negative SI or undersaturated scenarios as we do about the positive SI – and especially the higher risk areas SI > 1. So, the first step in this approach is a binary logistic XGBoost Classifier with logloss evaluation. The "watch zone" is defined as SI > 0, and SI ≤ 0 are labeled as "safe". A total of 3,500 lab samples were processed – yielding 350,000 total training samples while varying temperature and pressure from atmospheric to downhole conditions. Given that we expanded from 3500 samples to 350,000 - Special care was taken to ensure that data leakage was avoided.

Table 3—Feature parameter space for Celestite SI estimation

Statistic	Temperature (°F)	Pressure (psi)	SR (mg/L)	SO ₄ (mg/L)	SG
Mean	198.14	1717.02	103.46	1906.68	1.0403
Std Dev	72.85	1024.75	113.57	599.77	0.0155
Min	77.00	15.00	0.839	2.00	1.0004
Max	325.00	3500.00	931.73	7341.79	1.1723

Table 4—Optimized hyperparameters for XGBoost Binary Classifier model - "Safe" SI ≤0, and "Watch" SI > 0

Parameter	Value
learning_rate	0.072
max_depth	5
subsample	0.746
colsample_bytree	0.905
n_estimators	300
reg_lambda	2.574

Classification results indicate that the model - given the raw features of sg, temperature, pressure, SO₄, and Sr – is reliably good at filtering out the "safe" zones. A macro F-1 score of 0.995 tells us that both classes have extremely high precision and recall – indicated by only 38 false Safe labels out of 25894 total in the test dataset. AUC of 0.997 indicates excellent class separability.

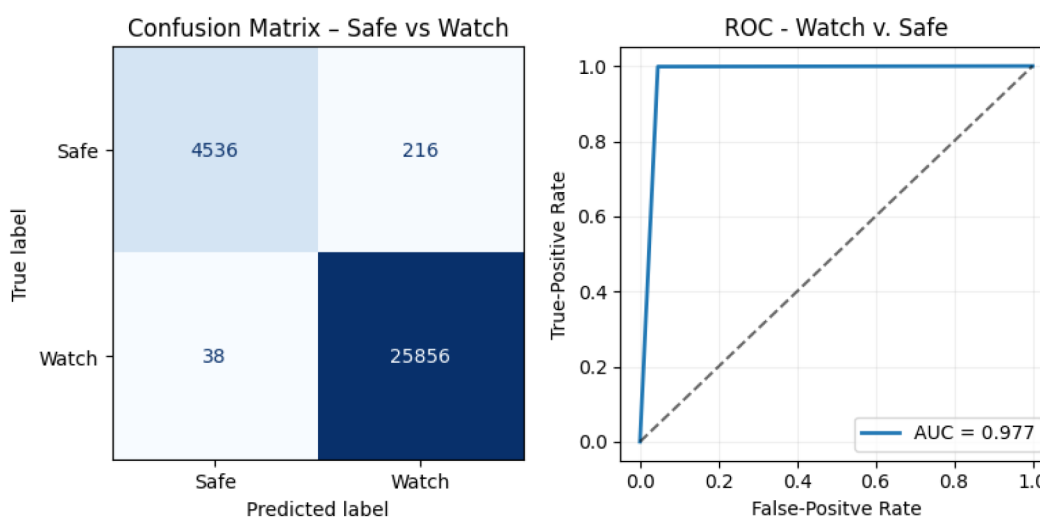


Figure 8—Binary classification results in first stage of Celestite SI estimator. Confusion matrix (left) and Receiver-Operating-Characteristic curve with Area-Under-Curve metric(right).

Next an XGBoost regressor is trained on the "Watch" zone samples. The raw parameters were passed into the model – and features such as $\text{Temperature} * \log(\text{Sr} * \text{SO}_4)$ were built into the model. A tree-based regressor was preferred for this model primarily to maximize explainability of results. It's understood that ultimately AI Agents needed not only to estimate SI values but provide users with an explanation as to why the value was selected. Tree-based regressors such as XGBoost allow for SHAP explainers – which are easily consumed by AI Agents – and explained clearly to users.

Table 5—Optimized hyperparameters for XGBoost Regressor model - "Watch" zone

Parameter	Value
learning_rate	0.054
max_depth	3
subsample	0.712
colsample_bytree	0.679
n_estimators	1050
reg_lambda	4.828
reg_alpha	1.358

Prediction results for values in the "watch" zone ($\text{SI} > 0$) are shown in Figure 9, in an actual vs predicted plot for SI. This shows how predictions are generally accurate (close to the plot diagonal) but tend to have a larger variance at values of $\text{SI} < 0.5$. Note that the results tend to cluster on the upper half of the plane, corresponding to a model prediction that is more conservative (higher predicted SI) than the real SI value. This behavior is expected, as we are considering only the concentration of Sr and SO_4 – while there are many more competing interactions between the other ions present in produced water. This picture is supported by Figure XY, where the residuals ($\text{SI predicted} - \text{true SI}$) are shown, highlighting the accuracy of the ML

model especially for $SI > 0.5$, where errors are typically below ± 0.05 SI units, up to the very high SI values ($SI > 1$) where accuracy is remarkable: within ± 0.01 SI units.

Table 6—Results for XGBoost Regressor model - "Watch" zone

Metric	SI
MAE	0.014
RMSE	0.034
R^2	0.952
Accuracy	97.3% (within $\pm 0.05 \log_{10}$)

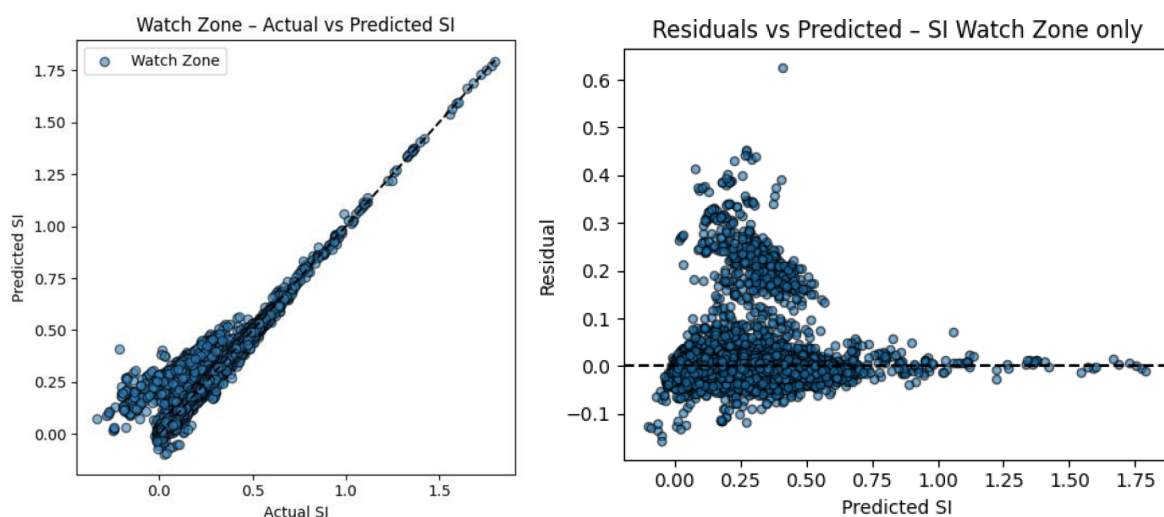


Figure 9—Regression results for watch zone samples (left) and model prediction residuals (right) – Celestite SI

These results, provide confidence over the model and framework as it can provide users with an accurate picture of the thermo-chemical risks at any temperature and pressure encountered in the reservoir. Note here that this model is easily scalable as once the space of the chemical variables (concentration of each ion) is earned by the machine learning model, this can be applied with minimal tuning to other reservoirs.

SI evaluation beyond water chemistry data

The typical inorganic scale risk evaluation workflow relies entirely on the thermochemistry of the system. Scaling index calculations address only the thermodynamics of the system, and do not include any kinetic evaluation for the Celestite precipitation and deposition process. It is known, for example, that the dynamic nature of fluid flow in oilfield operations strongly influences the kinetics of scaling and therefore modelling the produced fluid flow dynamics is required in order to accurately model scale deposition [17] [1]. These parameters include the presence of turbulence, which can influence the rate of mass transfer to surfaces and alter localized saturation conditions, and flow dynamics, which dictate shear forces and residence times within the wellbore [18]. The integration of such factors, beyond simple thermodynamic considerations, offers a more comprehensive and accurate prediction of scaling indexes and enables proactive mitigation strategies [19]. Turbulence can reduce the thickness of the diffusion boundary layer, thereby accelerating

the transport of scaling ions to the pipe wall [4]. It has been recently shown that the critical supersaturation required for nucleation and subsequent scale growth, particularly for sulfate scales [20] [21]. As well, the presence of specific completion elements such as downhole restrictions, can locally enhance turbulence and significantly influence scale deposition rates and morphology [22]. In particular, inflow control valves (ICV) and other perforated completion elements were shown to play a paramount role in accelerating deposition by introducing areas of high liquid shear stress, where scale tends to accumulate. However, conventional scale prediction models often simplify fluid dynamics by assuming static conditions, which can lead to inaccuracies in complex downhole environments [23].

To address these limitations, an advanced scale data analysis was developed aiming to expose causal relations between Celestite scale deposition and wells parameters with greater precision by considering factors beyond just thermochemical risk, such as flow regime, geometry, hole angle, and completion string details. Analyzing these complex interactions requires a robust data model and a platform, which can integrate temporal and static features, including physics-based simulator results (flow conditions, temperature, pressure) and lab chemical data, augmented by machine learning. This approach facilitates a more scalable and comprehensive solution to scale prediction. Timeseries production data, including fluid rates, water cut, temperature, and pressure, alongside static well parameters like completion design and deviation surveys, were used as inputs for the ML model together with 64 features obtained from Nodal simulations. Simulations were performed to match historical production data since the moment at which the water cut reached 2% and at a frequency of one per month of production up to a total of 25 years of production history. A total of 10 wells with 70 documented scale events were selected for this exploratory study for a grand total of 3000 simulations. The multivariate time series representing a section of depth (100 ft each, approximately 100 sections per well) was then considered as a separate dataset for the simulation.

The analysis incorporated static well construction features such as deviation angle, tortuosity, tubing diameter, and their distance from scale discoveries, as well as dynamic features related to fluid flow. A summary of the correlation between these data and scale events is shown in [Figure 10](#).

Looking at static well features first; from the figure it appears that most scale events were registered at well sections with high inclination angle (between 40 and 60 deg). It is indeed known that inclined wells often exhibit more complex flow regimes, including stratified or slug flow, which can contribute to localized variations in fluid velocity and shear stress, thereby promoting scale deposition [27]. Remarkably, a strong correlation between scaling and the presence of an inner diameter (ID) tubing change was observed, possibly at points of increased turbulence due to abrupt changes in flow area [18]. This suggests that such geometric discontinuities induce localized pressure drops and turbulent eddies, which can significantly enhance the rate of mineral precipitation on the pipe walls [28]. Scale was observed immediately upstream of the change in diameter, as flow experienced a sudden expansion. While this may not be immediately connected with high localized turbulence, we speculate this could create a zone of reduced flow velocity or eddies, promoting increased residence time for supersaturated solutions and enhanced mineral deposition [29] [30], [31]. Moreover, the available well intervention and scaling reports provide an incomplete depiction of scale formation's actual extent, as detection relies on indirect methods or only occurs following production impairment. Typically, caliper-like gauges are employed, which halt their descent within the well at the uppermost scale occurrence, leaving deeper sections inaccessible. We therefore cannot exclude that sections of the tubing even more affected by scale could be present at deeper depths.

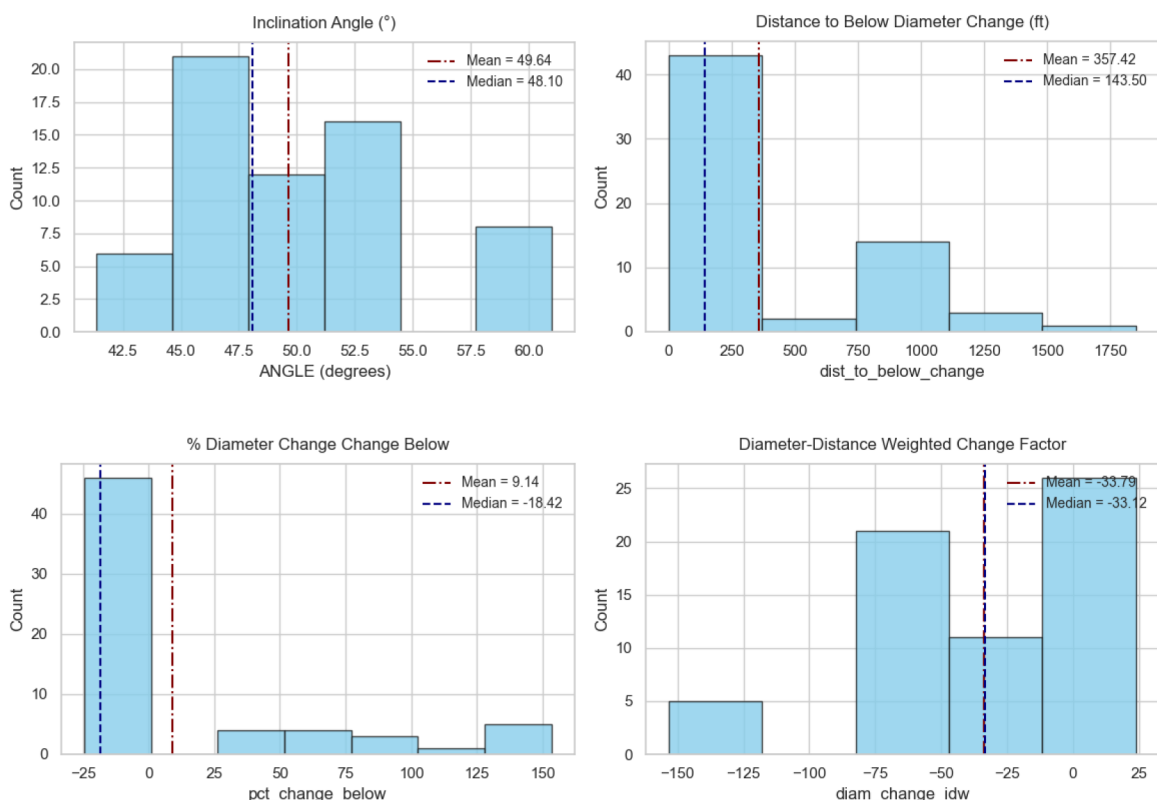


Figure 10—Static correlations from well angle (top-left), distance to a diameter change below scale discovery (top-right), magnitude (percentage) of change below scale discovery depth (bottom-left), and a distance-weighted diameter change factor (bottom-right)

Introducing fluid dynamic features in the analysis, it is also evident the correlation between scale and the fluids' Reynolds number (Re). Taking the example of the well shown in Figure 11, with several reported scaling events at timestamps between 2004-04 and 2024-02. By visualising both the calculated SI and the local Reynolds number computed with Prosper, it is clear that the conditions leading to the discovery of scale are characterized by both high turbulence and a $SI > 0.5$, shown by lighter colors in the plot. In agreement with the previous analysis, we note that the visible step in Re at around measured depth 9000 ft corresponds to a tubing ID change from 2.4" to 3.8". Such correlation was found in most of the reported scaling events and confirms the role of turbulence as a triggering force for the deposition of Celestite scale. This observation underscores the necessity of integrating fluid dynamic parameters into scale prediction models to accurately identify high-risk zones for deposition [32].

To integrate wells' static and hydrodynamic features into the thermochemical model for SI estimation, a machine learning workflow was developed. All data from well construction, production rates and Prosper simulations were fed into a comprehensive dataset for model training and validation. Several standard machine learning models were tested, including LSTM, transformers, neural ordinary differential equation (nODE) and autoencoders, with the goal of predicting the development of scale risk along the production history timeline and reproduce the increase in probability of Celestite deposition to the point of scale discovery (scale event).

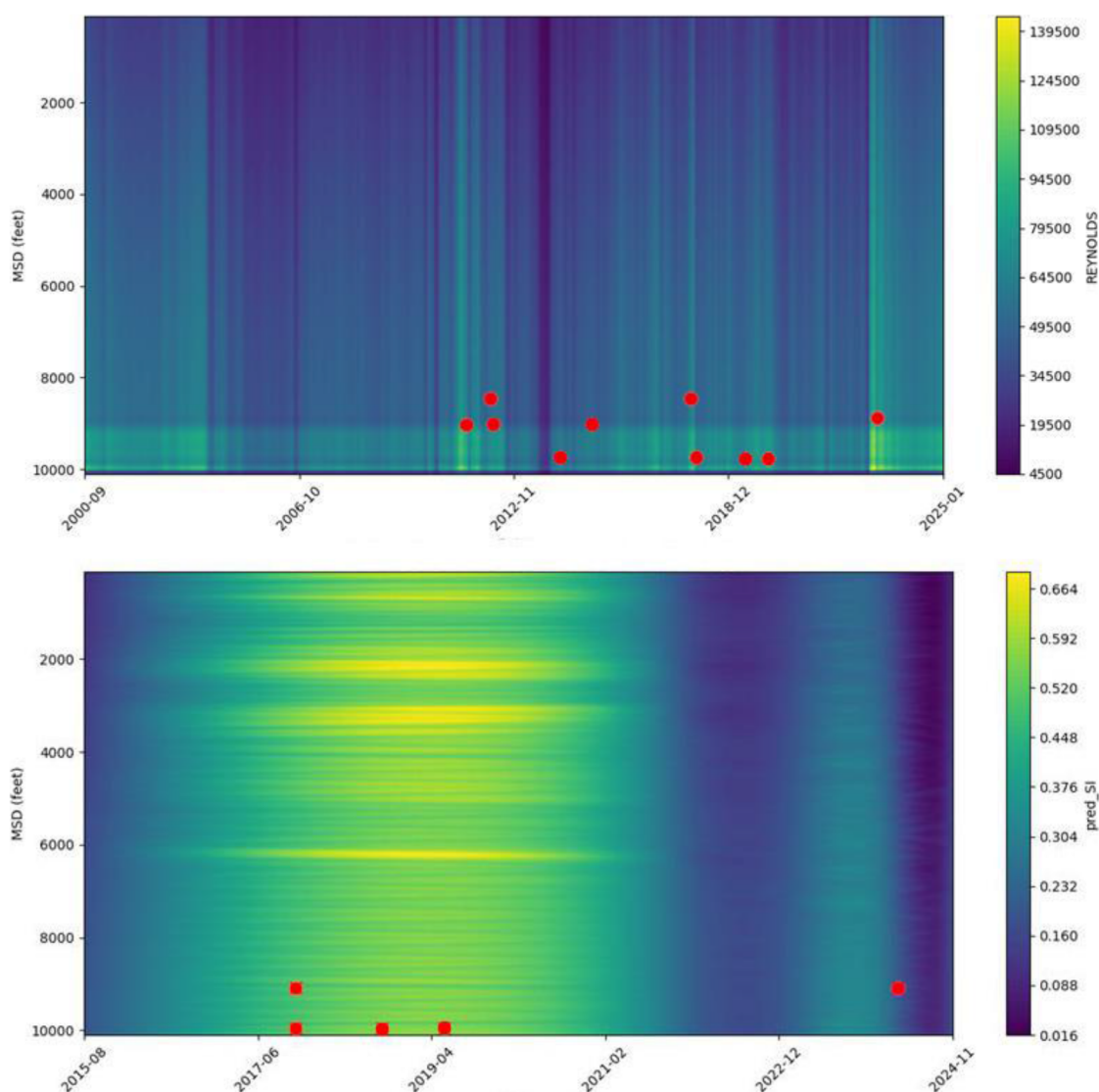


Figure 11—Dynamic correlations with scale discovery dates (red dots) – Reynolds number (top) and Celestite SI (bottom)

SiPro is a unified, model-agnostic pipeline for early celestite-scale prediction that standardizes data handling, labeling, training, and evaluation. Multivariate time series are harmonized (units/names), aligned by well–depth and sample month, and augmented with simple temporal features (lags/rolls, deltas) plus explicit missingness indicators; static descriptors are appended where available. Windows of length L stride S are constructed per entity, and targets are derived from a time-to-event scheme relative to the first confirmed scale discovery. Because timestamping and sampling are imperfect, we explored several label-smoothing approaches to reduce labeling noise and to make the targets consistent with early-warning use. Models are trained under identical settings (binary cross-entropy, AdamW, early stopping, gradient clipping), with optional post-hoc probability calibration and cumulative-max smoothing for stable risk trajectories.

Scale discoveries are extremely rare when compared to the life of a producing well. In order to mitigate the sparse nature of these events – and to provide the models with more domain-centric risk profiles, we evaluated several target-shaping policies: classic linear/quadratic/exponential/logistic decays; a temporal smoothing approach inspired by medical time-to-event labeling for atherosclerotic plaque progression (progressive risk as the event nears) [33]; and a custom WC-gated policy tailored to the production context. The custom label $y_i \in [0,1]$ uses a 3-month rolling water-cut W_i and gate threshold θ to define the first

gate time t_g (e.g., $W_t \geq \theta$ for k consecutive months). For the first event at t_e , the pre-event label increases monotonically through a steady ramp, a short surge, and a capped plateau; only t_e is 1. Post-event, the label tapers (post scale removal). A compact definition is

$$y_t = \begin{cases} 0, & t < t_g \\ f \frac{t - t_g + 1}{R}, & t_g \leq t < t_g + R, \quad R = \max\{0, (t_e - t_g) - S - P\}, \\ f + (c - f) \ell\left(\frac{t - (t_g - S - P)}{S}; \alpha\right), & t_g - S - P \leq t < t_g - P, \\ c, & t_g - P \leq t < t_e, \\ 1, & t = t_e, \\ f + (1 - f) * e^{-\lambda(t - t_e)}, & t > t_e, \end{cases}$$

where $f \in (0, 1)$ is the floor, $c \in (f, 1)$ the pre-event cap, S the surge length, P the plateau length, $\lambda > 0$ the taper rate, and $\ell(u; \alpha)$ is a normalized logistic mapping $u \in [0, 1]$ to $[0, 1]$ with steepness α . We enforce monotonicity on $[t_g, t_e]$ via a cumulative-max operator and allow the ramp to auto-size to the gate-to-event gap. This custom policy is exploratory and intended to encode domain intuition (WC gate $\rightarrow$ gradual scale accumulation $\rightarrow$ rapid pre-event rise with a sustained high-risk signal preceding the scale discovery). Substantial work remains to finalize a celestite-specific labeling strategy—e.g., alternative gating definitions, cost-aware shaping, multi-event handling, celestite scale deposition mechanisms and validation across fields and sampling regimes. The labeling techniques tested are summarized in figure 12.

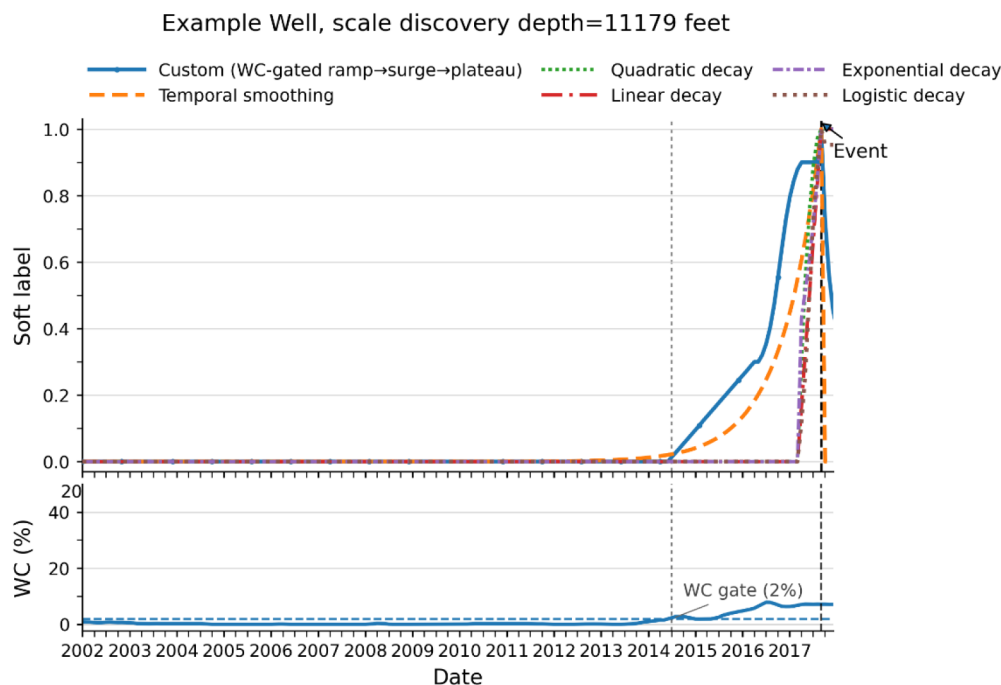


Figure 12—Labeling techniques tested to provide the ML models with a more domain intuitive risk profile

We compared multiple temporal model families under this identical pipeline (logistic regression, LSTM, TCN, Transformer, ODE-RNN/Latent-ODE) to understand which inductive biases best fit the data. To illustrate signal capture—not deployment—we present LSTM results as the strongest performer on this dataset: confusion matrix at a validation operating point (low false-positive rate, high specificity), ROC (AUC ≈ 0.87), calibrated probability quality, and per-well risk trajectories. These findings are preliminary but indicate that ML can surface physically consistent pre-event patterns in this setting. And perhaps equally as important – given that the requisite simulator results and source data can be integrated, ingested, and

available into a single platform (CDF) - deployment and scale-out of this type of model to a field or multiple fields is a realistic endeavor.

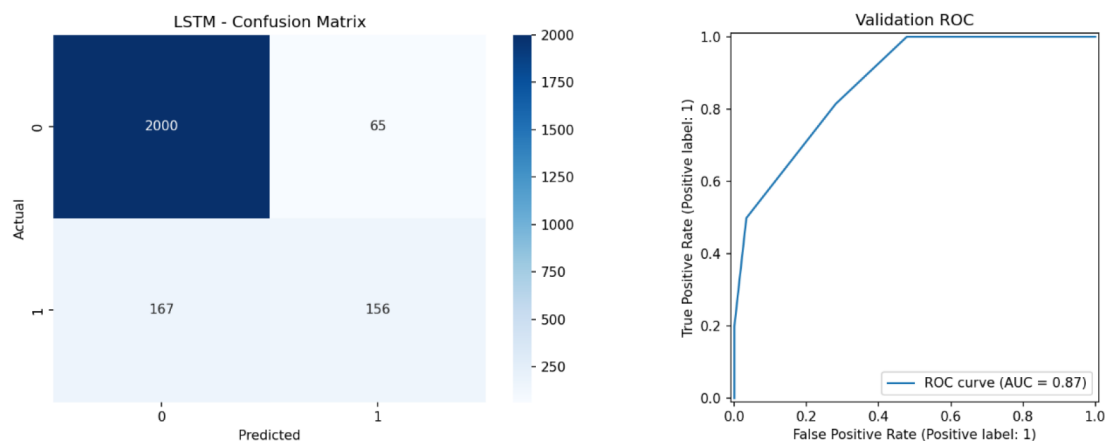


Figure 13—Classification results (validation windows) from the SiPro-LSTM model. Confusion matrix (left) and Receiver-Operating-Characteristic curve with Area-Under-Curve metric(right).

Results for two sample wells are shown in Figure 14. Among all tested models, LSTM was found to offer the best performance, accurately capturing the temporal dependencies inherent in the data and predicting an increasing probability of Celestite deposition in correspondence of recorded scaling events. Other models struggled to reproduce the subtle, long-term trends indicative of scale accumulation. The effectiveness of the LSTM model can be attributed to its proficiency in handling sequential data, which is crucial for forecasting time-dependent phenomena such as scale accumulation, by discerning intricate patterns that precede and accompany scale formation [31]. Yet qualitatively correct results were obtained only a few analysed wells. One possible source of this inaccuracy lies in the sparsity and scarcity of data. Throughout the long history of production of a well (up to 25 years), scaling events are only a few documented, leading to a limited number of positive labels for training the machine learning models. This limitation often necessitates advanced techniques like data augmentation or transfer learning to enhance model robustness [32]. Also, a major shortfall is the lack of data describing the progressive Celestite thickness growth on the tubing as, when found, scale is immediately removed. Another limitation arises from the simplistic approach of the current model, which treats scale events as discrete occurrences rather than a continuous, evolving process influenced by a multitude of interdependent parameters that change dynamically over time. This highlights the need for models that can infer continuous changes in scale thickness from sparse observations, potentially by incorporating physics-informed neural networks or inverse modeling techniques. A possible workaround to improve models in this direction is to continue work on field or reservoir-specific label-smoothing techniques which aim at introducing a synthetic signature of scale growth in the training data, such as a progressive (linear, exponential, etc.) raise in the scaling probability, building up to the measured event. Albeit somewhat arbitrary, similar approaches were found successful in supporting the identification of rare events in other fields such as healthcare [36]. Further development of a field or reservoir specific custom Celestite scale development risk label is merited - and can help to further enforce the monotonic correlations observed in the analysis of the SiPro data set, and impact local domain expertise onto the models.

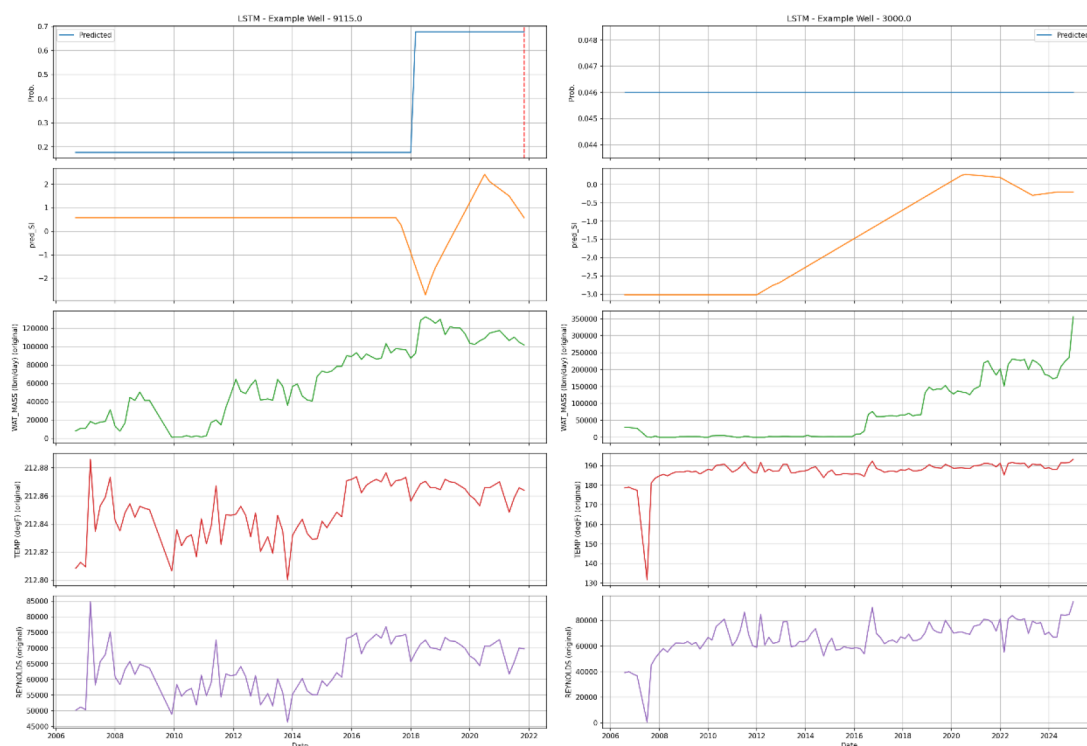


Figure 14—Early warning signals for two depths of study in two example wells – only dynamic features plotted (probability, SI, water mass flow rate, temperature, and Reynolds number). Left: Example scale discovery event shows a sharp rise well in advance of scale discovery – despite mixed chemical indicators. Right: maintains near-zero probability, correctly indicating no event at 3000 feet (above typical Celestite formation depth) - despite rise in SI trend.

Conclusions

This work tackled the persistent challenge of scale risk management in offshore oilfields, where limited access to comprehensive water chemistry data hinders proactive mitigation of mineral scale formation. Traditional approaches rely on time-consuming and costly laboratory analyses and thermodynamic simulations, which delay decision-making and often fail to capture dynamic subsurface conditions. The inability to continuously assess scaling risk, especially under seawater breakthrough scenarios, presents a significant operational and economic burden.

Several approaches have been presented. First is the seawater percentage estimation using machine learning. With only specific gravity as input parameter, the ML approach achieved an accuracy of 95.83% and 96.30% for reservoirs A and B, respectively, compared to the current operator's formula which yields only 86.80% and 90.84%.

Second, a machine learning model was developed for reconstructing ion concentrations using SG as input. These predicted concentrations are then used to estimate the scale index. The model demonstrated strong performance for key scale-relevant ions: chloride and salinity yielded accuracies of 97.96% and 97.26%, respectively, followed by sodium at 96.93%. Although strontium and sulfate showed slightly lower accuracies (88.21% and 84.53%), they still reflect acceptable predictive power given the limited input data. Magnesium prediction posed the highest error with 82.35% accuracy.

Third, an ML-based model was trained on a simulated dataset for Celestite Saturation Index (SI) prediction using sparse chemical data as input. The XGBoost model demonstrated strong performance in the "watch" zone ($SI > 0$), achieving a mean absolute error of 0.014 and 97.3% of predictions falling within $\pm 0.05 \log_{10}$ units. This highlights the model's potential for reliable SI estimation without requiring full thermodynamic simulations.

We also developed SiPro, a unified sequence-modeling pipeline that converts multivariate operating and chemistry historical data into calibrated early-warning probabilities of first celestite discovery. Targets

are time-to-event labels (with several smoothing options explored, including a WC-gated custom policy) so risk rises as seawater breakthrough progresses. Under identical preprocessing-splitting-training across model families, the LSTM performed best—ROC-AUC ≈ 0.87 with low false-positive rates and per-well trajectories that increase ahead of events and co-vary with static and dynamic features. These findings are preliminary but show ML can recover physically consistent precursors. Next steps are to expand training data, refine data labeling, and quantify lead time and intervention value, enabling downstream integration.

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